

State Ohm's Law and Kirchhoff's Laws (KCL & KVL)

Ohm's Law

Statement:

At constant temperature, the current through a conductor is directly proportional to the applied voltage.

$$V = IR$$

In DC circuits:

Used to determine unknown current, voltage, or resistance in linear resistive DC networks.

Kirchhoff's Current Law (KCL)

Statement:

The algebraic sum of currents at any node in a DC circuit is zero.

$$\sum I = 0$$

Meaning:

Total current entering a node equals total current leaving the node.

(Based on conservation of charge)

Kirchhoff's Voltage Law (KVL)

Statement:

The algebraic sum of all voltages around any closed loop in a DC circuit is zero.

$$\sum V = 0$$

Meaning:

Sum of voltage rises equals sum of voltage drops.

(Based on conservation of energy)

2. Define the Following Terms

- **Branch:**
A part of a circuit containing a single electrical element between two nodes.
 - **Node:**
A point where two or more circuit elements are connected.
 - **Junction:**
A node at which three or more branches meet.
 - **Loop:**
Any closed path in a circuit.
 - **Mesh:**
A loop that does not contain any other loop inside it.
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3. Comparison of Resistive Series and Parallel Circuits

Parameter	Series Circuit	Parallel Circuit
Current	Same through all resistors	Divides among branches
Voltage	Divides across resistors	Same across all resistors
Equivalent Resistance	$R_{eq} = R_1 + R_2 + \dots$	$\frac{1}{R_{eq}} = \sum \frac{1}{R}$
Reliability	Low (failure breaks circuit)	High (other paths exist)
Application	Voltage divider	Current divider

1. Electric Current (I)

Definition:

Electric current is the **rate of flow of electric charge** through a conductor.

Mathematical form:

$$I = \frac{Q}{t}$$

SI unit: Ampere (A)

2. Voltage (V)

Definition:

Voltage is the **electric potential difference** between two points, i.e., the **work done per unit charge** to move a charge from one point to another.

Mathematical form:

$$V = \frac{W}{Q}$$

SI unit: Volt (V)

Think of it as:

The **push** that makes current flow. No push, no party.

3. Electric Power (P)

Definition:

Electric power is the **rate at which electrical energy is consumed or converted** in a circuit.

Mathematical forms:

$$P = VI$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

SI unit: Watt (W)

Reality check:

1.2 THEORY QUESTIONS (IN DETAIL)

1 Star-Delta (Y-Δ) Transformation (Detailed Derivation)

Why Star-Delta is Needed

In many DC networks, resistors are neither purely series nor purely parallel. Star-Delta transformation helps convert such networks into solvable forms **without changing terminal behavior**.

(a) Star (Y) to Delta (Δ) Transformation

Star Network

Let a star network have three resistances:

- R_A between terminal A and center O
 - R_B between terminal B and center O
 - R_C between terminal C and center O
-

Step 1: Resistance between terminals A and B in Star

Current flows through R_A and R_B in series.

$$R_{AB}^{(Star)} = R_A + R_B$$

Step 2: Resistance between A and B in Delta

In delta, between A and B:

- Direct resistor R_{AB}
- Alternate path through $R_{AC} + R_{CB}$

Equivalent resistance:

$$R_{AB}^{(Delta)} = \frac{R_{AB}(R_{AC} + R_{CB})}{R_{AB} + R_{AC} + R_{CB}}$$

Step 3: Equate Star and Delta Resistances

$$R_A + R_B = \frac{R_{AB}(R_{AC} + R_{CB})}{R_{AB} + R_{BC} + R_{CA}}$$

Solving similarly for other terminal pairs and simplifying:

✓ Final Star → Delta Formula

$$R_{AB} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

$$R_{BC} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A}$$

$$R_{CA} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}$$

(b) Delta (Δ) to Star (Y) Transformation

Delta Network

Let delta resistances be:

- R_{AB}, R_{BC}, R_{CA}

Step 1: Resistance between A and B in Delta

$$R_{AB}^{(Delta)} = R_{AB} \parallel (R_{BC} + R_{CA})$$

Step 2: Resistance between A and B in Star

$$R_{AB}^{(Star)} = R_A + R_B$$

Step 3: Equate and Solve

By solving all three terminal equations:

✓ Final Delta → Star Formula

$$R_A = \frac{R_{AB}R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$
$$R_B = \frac{R_{AB}R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$
$$R_C = \frac{R_{BC}R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

Key Point to Write

Star–Delta transformation preserves resistance between any two terminals.

2 Network Theorems (Fully Explained)

(a) Thevenin's Theorem (In Detail)

Statement

Any linear, bilateral DC network containing sources and resistances can be replaced by:

- A single voltage source V_{th}
- In series with **equivalent resistance** R_{th}
as seen from the load terminals.

Explanation

Instead of solving the entire circuit repeatedly for different loads, the network is simplified into a **Thevenin equivalent**, making analysis easy.

Steps

1. Remove load resistance R_L
2. Find open-circuit voltage → V_{th}

3. Deactivate sources

- Voltage source → short circuit
- Current source → open circuit

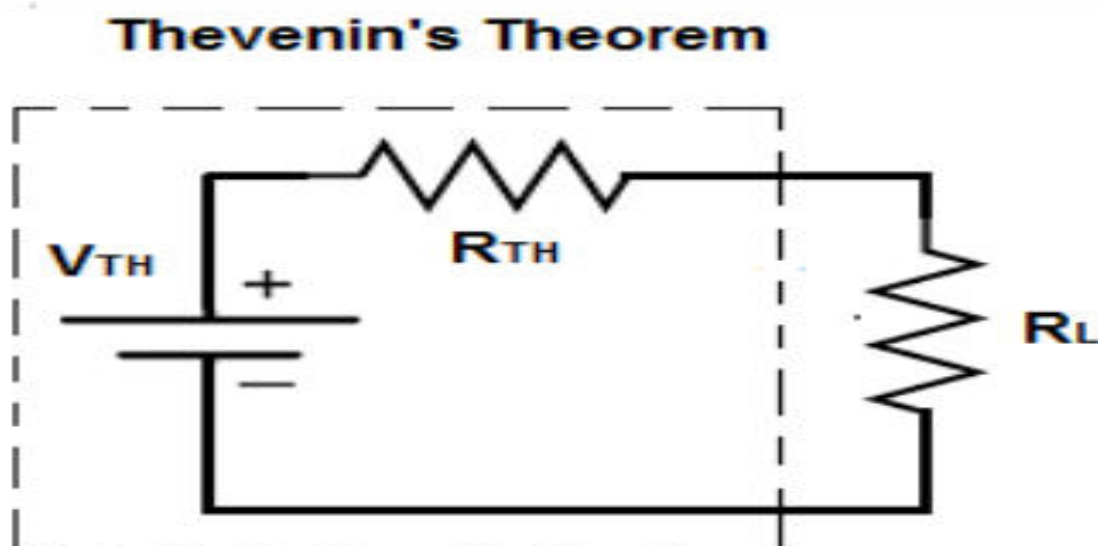
4. Find equivalent resistance → R_{th}

5. Reconnect load and find current:

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

Application

Widely used in power systems and circuit optimization.



(b) Norton's Theorem (In Detail)

Statement

Any linear DC network can be replaced by:

- A **current source** I_N
- In parallel with **resistance** R_N

Relation with Thevenin

$$I_N = \frac{V_{th}}{R_{th}}, R_N = R_{th}$$

Steps

1. Short the load terminals
 2. Find short-circuit current $\rightarrow I_N$
 3. Find equivalent resistance $\rightarrow R_N$
-

Use

Preferred for circuits with parallel branches and current calculations.

(c) Superposition Theorem (In Detail)

Statement

In a linear circuit with multiple independent sources, the response in any element equals the algebraic sum of responses due to each source acting alone.

Procedure

1. Consider one source at a time
 2. Replace other sources:
 - Voltage source $\rightarrow$ short
 - Current source $\rightarrow$ open
 3. Calculate response
 4. Add all responses algebraically
-

Limitation

✗ Cannot be applied directly to power calculation.

3 Circuit Analysis Using KCL / KVL (Stepwise)

KCL (Node Voltage Method)

Steps

1. Choose reference (ground) node
2. Assign node voltages
3. Apply KCL:

$$\sum I_{entering} = \sum I_{leaving}$$

4. Express currents using Ohm's law
 5. Solve simultaneous equations
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KVL (Mesh/Loop Method)

Steps

1. Choose loop direction
2. Apply KVL:

$$\sum V = 0$$

3. Write voltage drops using IR
 4. Solve loop equations
-

Star Circuit Application

Apply KCL at central node → write current equations → find branch currents.

4 Transient Analysis (Very Detailed)

(a) RC Circuit – Capacitor Charging

Circuit Behavior

- Initially capacitor acts as **short circuit**
 - Finally behaves as **open circuit**
-

Applying KVL

$$V = iR + v_c$$

But,

$$i = C \frac{dv_c}{dt}$$

Substitute:

$$V = RC \frac{dv_c}{dt} + v_c$$

Solving Differential Equation

$$\frac{dv_c}{V - v_c} = \frac{dt}{RC}$$

Integrating:

$$v_c(t) = V(1 - e^{-t/RC})$$

Current Equation

$$i(t) = \frac{V}{R} e^{-t/RC}$$

(b) RL Circuit – Inductor Charging

Circuit Behavior

- Inductor initially behaves as **open circuit**
 - Finally behaves as **short circuit**
-

Applying KVL

$$V = iR + L \frac{di}{dt}$$

Rearranging:

$$\frac{di}{V/R - i} = \frac{R}{L} dt$$

Solution

$$i(t) = \frac{V}{R}(1 - e^{-tR/L})$$
$$v_L(t) = Ve^{-tR/L}$$

Time Constant Interpretation

- RC $\rightarrow \tau = RC$
- RL $\rightarrow \tau = \frac{L}{R}$

At $t = \tau$:

✓ 63.2% of final value reached

2.1 Important Definitions

1 Basic AC Quantities

Amplitude

The **maximum value** of an alternating quantity (voltage or current) measured from the zero reference level.

$$\text{Amplitude} = V_m \text{ or } I_m$$

Frequency

The **number of cycles** completed per second by an AC waveform.

$$f = \frac{1}{T} \text{ (Hz)}$$

Time Period

The **time taken** to complete one full cycle of an AC waveform.

$$T = \frac{1}{f} \text{ (seconds)}$$

Instantaneous Value

The value of an alternating quantity at **any particular instant of time**.

$$v(t) = V_m \sin(\omega t)$$

2 RMS, Average, Form & Peak Values

RMS (Root Mean Square) Value

The **effective value** of AC which produces the same heating effect as DC.

$$V_{rms} = \frac{V_m}{\sqrt{2}}, I_{rms} = \frac{I_m}{\sqrt{2}}$$

Average Value

The **mean value** of an AC waveform over **half cycle**.

$$V_{avg} = \frac{2V_m}{\pi}$$

Form Factor

The ratio of RMS value to average value of an AC waveform.

$$\text{Form Factor} = \frac{V_{rms}}{V_{avg}} = 1.11$$

Peak (Crest) Factor

The ratio of maximum value to RMS value.

$$\text{Peak Factor} = \frac{V_m}{V_{rms}} = 1.414$$

CYCLE : a **cycle** means **one complete repetition of a motion, wave, or process**, after which the system comes back to the **same position and state** it started from.

3 Power in AC Circuits

Active Power (True Power)

The power **actually consumed** by the circuit to do useful work.

$$P = VI \cos \phi (\text{Watts})$$

Reactive Power

The power that **oscillates** between source and reactive elements (L and C).

$$Q = VI \sin \phi (\text{VAR})$$

Apparent Power

The product of RMS voltage and RMS current.

$$S = VI(\text{VA})$$

Power Factor

The cosine of the phase angle between voltage and current.

$$\text{Power Factor} = \cos \phi = \frac{P}{S}$$

→ High power factor = better efficiency

4 Q-Factor and Bandwidth

Q-Factor (Quality Factor)

A measure of **sharpness of resonance** in an AC circuit.

$$Q = \frac{f_r}{BW}$$

Bandwidth

The range of frequencies between **upper and lower half-power frequencies**.

$$BW = f_2 - f_1$$

Relation

- High Q → Narrow bandwidth
- Low Q → Wide bandwidth

MODULE 2: AC CIRCUITS

2.2 THEORY QUESTIONS – VERY DETAILED

1 PURE AC CIRCUITS

(A) PURELY INDUCTIVE CIRCUIT

Circuit Description

A purely inductive circuit consists of an **ideal inductor L** connected across an AC supply. There is **no resistance**, hence **no real power loss**.

Applied Voltage

$$v(t) = V_m \sin \omega t$$

Basic Inductor Law

For an inductor,

$$v = L \frac{di}{dt}$$

This means **voltage across an inductor is proportional to the rate of change of current**.

Mathematical Derivation

$$V_m \sin \omega t = L \frac{di}{dt}$$
$$\frac{di}{dt} = \frac{V_m}{L} \sin \omega t$$

Integrating with respect to time,

$$i = -\frac{V_m}{\omega L} \cos \omega t$$

Using trigonometric identity,

$$\cos \omega t = \sin (\omega t + 90^\circ)$$

$$i = \frac{V_m}{\omega L} \sin (\omega t - 90^\circ)$$

✓ Phase Relation

- Current **lags voltage by 90°**
 - This lag occurs because the inductor **opposes change in current**
-

Inductive Reactance

$$X_L = \omega L$$

$$I_m = \frac{V_m}{X_L}$$

Power in Inductive Circuit

Instantaneous power:

$$p(t) = v(t)i(t)$$

$$p(t) = V_m I_m \sin \omega t \sin (\omega t - 90^\circ)$$

Average power over one cycle:

$$P = VI \cos \phi$$

For pure inductor:

$$\phi = 90^\circ \Rightarrow P = 0$$

Physical Meaning

- Energy is stored in the magnetic field during one half-cycle
- Returned to the source in the next half-cycle

- No energy is consumed
-

(B) PURELY CAPACITIVE CIRCUIT

Circuit Description

A purely capacitive circuit consists of an **ideal capacitor C** connected to an AC supply.

Applied Voltage

$$v(t) = V_m \sin \omega t$$

Capacitor Law

$$i = C \frac{dv}{dt}$$

This shows **current depends on rate of change of voltage**.

Mathematical Derivation

$$\begin{aligned} i &= C \frac{d}{dt} (V_m \sin \omega t) \\ i &= \omega C V_m \cos \omega t \end{aligned}$$

Using identity,

$$\begin{aligned} \cos \omega t &= \sin (\omega t + 90^\circ) \\ i &= \omega C V_m \sin (\omega t + 90^\circ) \end{aligned}$$

Phase Relation

- Current **leads voltage by 90°**
 - Because capacitor allows current to flow before voltage builds up
-

Capacitive Reactance

$$X_C = \frac{1}{\omega C}$$
$$I_m = \frac{V_m}{X_C}$$

Average Power

$$P = VI \cos 90^\circ = 0$$

Physical Meaning

- Energy stored in electric field
 - Energy returned to source
 - **No net power consumption**
-

2 RESONANCE IN AC CIRCUITS

Definition

Resonance occurs in an RLC circuit when:

$$X_L = X_C$$

Resonant Frequency

$$\omega_r = \frac{1}{\sqrt{LC}}$$
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

SERIES RESONANCE

Circuit

R, L, and C connected in series.

At Resonance

- $X_L = X_C$
- Net reactance = 0
- Impedance is minimum:

$$Z = R$$

Characteristics of Series Resonance

1. Circuit behaves as purely resistive
 2. Current is maximum
 3. Power factor is unity
 4. Voltage across L and C is very high
 5. Circuit is frequency selective
 6. Used in tuning and radio circuits
-

PARALLEL RESONANCE

- Impedance is maximum
 - Line current is minimum
 - Acts as rejector circuit
-

Comparison: Series vs Parallel Resonance

Aspect	Series	Parallel
Impedance	Minimum	Maximum

Aspect	Series	Parallel
Current	Maximum	Minimum
Application	Tuning	Filtering

3 PHASOR DIAGRAMS

(A) R–L SERIES CIRCUIT

Voltage Components

- $V_R = IR \rightarrow$ in phase with current
 - $V_L = IX_L \rightarrow$ leads current by 90°
-

Impedance Triangle

$$Z = \sqrt{R^2 + X_L^2}$$

Voltage Triangle

$$V = \sqrt{V_R^2 + V_L^2}$$

Power Triangle

- Active Power $P = VI \cos \phi$
 - Reactive Power $Q = VI \sin \phi$
 - Apparent Power $S = VI$
-

(B) R–C SERIES CIRCUIT

- Current leads voltage
- $V_C = IX_C$ lags current by 90°

$$Z = \sqrt{R^2 + X_C^2}$$

4 THREE-PHASE RELATIONS

(A) STAR (Y) CONNECTION

Voltage Relation

$$V_L = \sqrt{3}V_{ph}$$

Current Relation

$$I_L = I_{ph}$$

Derivation Idea

Line voltage is the **vector difference** of two phase voltages separated by 120° .

(B) DELTA (Δ) CONNECTION

Voltage Relation

$$V_L = V_{ph}$$

Current Relation

$$I_L = \sqrt{3}I_{ph}$$

Three-Phase Power

$$P = \sqrt{3}V_L I_L \cos \phi$$

Series RLC Circuit – Analysis

Circuit description

A resistor (R), inductor (L), and capacitor (C) are connected in series to an AC supply of angular frequency ω .

Same current I flows through R, L, and C (that's the whole point of series).

Step 1: Individual Impedances

- Resistor:

$$Z_R = R$$

- Inductor:

$$Z_L = j\omega L$$

- Capacitor:

$$Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

Step 2: Total Impedance of Series RLC Circuit

Since impedances add in series:

$$Z = Z_R + Z_L + Z_C$$

$$Z = R + j\omega L - \frac{j}{\omega C}$$

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

Magnitude of Impedance

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

This is the impedance expression (★ very important).

Step 3: Phase Angle (ϕ)

$$\tan \phi = \frac{\text{Net reactance}}{R}$$

$$\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R}$$

- If $\omega L > \frac{1}{\omega C} \rightarrow$ Inductive (current lags)
- If $\omega L < \frac{1}{\omega C} \rightarrow$ Capacitive (current leads)
- If equal $\rightarrow$ Resonance

Step 4: Power Factor of Series RLC Circuit

Definition

Power factor is the cosine of the phase angle between voltage and current.

$$\text{Power Factor} = \cos \phi$$

Expression for Power Factor

$$\cos \phi = \frac{R}{|Z|}$$

$$\cos \phi = \frac{R}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

Special Case: Resonance (Exam Favourite)

At resonance:

$$\omega L = \frac{1}{\omega C}$$

Then:

$$Z = R$$

$$\cos \phi = 1$$

a) Transient Response of RC Circuit (Charging)

[7 Marks Answer]

Definition

The transient response of an RC circuit is the temporary behavior of current and voltage when a DC voltage is suddenly applied, before the circuit reaches steady state.

Circuit Description

A resistor (R) and capacitor (C) are connected in series with a DC source V through a switch.

At $t = 0$, the switch is closed and the capacitor starts charging.

Working Principle

- Initially, capacitor behaves like a short circuit
- Current is maximum at $t = 0$
- As capacitor charges, current decreases
- Finally, capacitor acts as an open circuit

Derivation of Charging Equation

Applying KVL:

$$V = iR + \frac{q}{C}$$

Since $i = \frac{dq}{dt}$:

$$V = R \frac{dq}{dt} + \frac{q}{C}$$

Solving the differential equation:

Capacitor Voltage

$$V_C(t) = V(1 - e^{-\frac{t}{RC}})$$

Charging Current

$$i(t) = \frac{V}{R} e^{-\frac{t}{RC}}$$

Time Constant

$$\tau = RC$$

- At $t = \tau$: capacitor charges to 63.2% of final voltage
- At $t = 5\tau$: capacitor is fully charged

Waveforms (Description)

- Capacitor voltage V_C rises exponentially from 0 to V
- Current i falls exponentially from maximum to zero

 Voltage curve: rising exponential

 Current curve: decaying exponential

Conclusion

During charging, the RC circuit exhibits exponential transient behavior governed by the time constant RC .

(b) Transient Response of RL Circuit (Current Growth)

[7 Marks Answer]

Definition

The transient response of an RL circuit describes the growth of current when a DC voltage is suddenly applied.

Circuit Description

A resistor (R) and inductor (L) are connected in series with a DC source V through a switch.

At $t = 0$, the switch is closed.

Working Principle

- Inductor initially opposes change in current
 - Initial current is zero
 - Current gradually increases
 - Finally reaches steady value $\frac{V}{R}$
-

Derivation of Current Growth Equation

Applying KVL:

$$V = iR + L \frac{di}{dt}$$

Rearranging:

$$\frac{di}{dt} + \frac{R}{L} i = \frac{V}{L}$$

Solving the differential equation:

Current Growth Equation

$$i(t) = \frac{V}{R} (1 - e^{-\frac{t}{\tau}})$$

Where:

$$\tau = \frac{L}{R}$$

Time Constant

- At $t = \tau$: current reaches 63.2% of final value
- At $t = 5\tau$: steady state reached

Waveform (Description)

- Current rises exponentially from 0 to $\frac{V}{R}$
- Inductor voltage decays exponentially

 Current waveform: rising exponential

Conclusion

The RL circuit shows exponential current growth, controlled by the time constant $\frac{L}{R}$.

5 TWO WATTMETER METHOD (EXTREMELY IMPORTANT)

Purpose

Measures **total power** in a **3-phase, 3-wire system**.

Working Principle

Each wattmeter measures power in one phase.

Total power is algebraic sum.

$$P = W_1 + W_2$$

Power Factor Expression

$$\tan \phi = \frac{\sqrt{3}(W_1 - W_2)}{W_1 + W_2}$$

Special Cases

Power Factor Wattmeter Readings

Unity	$W_1 = W_2$
0.5	One reads zero
< 0.5	One negative
Zero	Equal and opposite

Merits

1. Only two wattmeters required
 2. Suitable for balanced & unbalanced loads
 3. Simple and economical
-

Demerits

1. Not accurate at low power factor
 2. Negative reading handling required
-

6 ADVANTAGES OF 3-PHASE SYSTEM

1. Constant power
2. Less conductor material
3. Higher efficiency
4. Smaller machines
5. Higher torque
6. Economical transmission
7. Better regulation

MODULE 3: MAGNETIC CIRCUITS & TRANSFORMERS

3.1 IMPORTANT DEFINITIONS

1 Magnetic Circuit Terms

Magnetic Flux (Φ)

The total number of magnetic lines of force passing through a magnetic circuit.

$$\Phi = BA$$

Unit: Weber (Wb)

Magnetic Flux Density (B)

The amount of magnetic flux per unit area normal to the flux.

$$B = \frac{\Phi}{A}$$

Unit: Tesla (T)

Magnetomotive Force (MMF)

The force that establishes magnetic flux in a magnetic circuit.

$$\text{MMF} = NI$$

Unit: Ampere-turns (AT)

Reluctance ($\mathcal{R}$)

The opposition offered by a magnetic circuit to the flow of magnetic flux.

$$\mathcal{R} = \frac{l}{\mu A}$$

Unit: AT/Wb

Permeability (μ)

The ability of a material to conduct magnetic flux.

$$\mu = \mu_0 \mu_r$$

Unit: H/m

Magnetic Field Strength (H)

The magnetizing force per unit length of the magnetic circuit.

$$H = \frac{NI}{l}$$

Unit: A/m

2 Transformer Terms

Transformer

A **static electrical device** that transfers AC power from one circuit to another at the **same frequency** but usually at different voltage levels through **mutual induction**.

Voltage Regulation of Transformer

The change in secondary terminal voltage from **no-load to full-load**, expressed as a percentage of no-load voltage.

$$\% \text{Regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100$$

Efficiency of Transformer (η)

The ratio of output power to input power of a transformer.

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100$$

MODULE 3: MAGNETIC CIRCUITS & TRANSFORMERS

3.2 THEORY QUESTIONS (IN DETAIL)

1 Magnetic Circuits

(a) B–H Curve

Definition

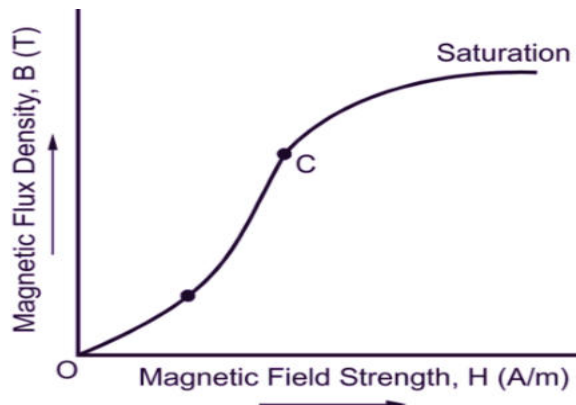
The B–H curve represents the relationship between **magnetic flux density (B)** and **magnetic field strength (H)** of a magnetic material.

Explanation

- Initially, B increases linearly with H
 - As H increases further, B increases slowly
 - Finally, material reaches **saturation**
-
-

Importance

- Helps select magnetic materials
- Indicates saturation point
- Used in transformer core design



(b) Hysteresis Loop

Definition

The closed B – H curve obtained when a magnetic material is subjected to **cyclic magnetization** is called the hysteresis loop.

Features

- **Retentivity:** Residual flux density when $H = 0$
 - **Coercivity:** Reverse field required to reduce flux to zero
 - **Loop Area:** Energy loss per cycle
-

Significance

- Narrow loop $\rightarrow$ transformer cores
 - Wide loop $\rightarrow$ permanent magnets
-

(c) Hysteresis Loss

Cause

Due to repeated magnetization and demagnetization of core material.

Expression

$$P_h = \eta B_{max}^{1.6} fV$$

Reduction

- Use silicon steel
 - Operate at lower flux density
-
-

(d) Eddy Current Loss

Cause

Circulating currents induced in the core due to changing magnetic flux.

Expression

$$P_e = K_e B_{max}^2 f^2 t^2 V$$

Reduction

- Laminated core
 - High resistivity materials
-
-

2 Transformer Principle

(a) Working Principle

A transformer works on the principle of **mutual induction**.

When AC is applied to the primary winding:

- Alternating flux is produced
- Flux links secondary winding
- EMF is induced in secondary

(b) Construction of Single-Phase Transformer

Main Parts

1. **Magnetic Core** – laminated silicon steel
 2. **Primary Winding**
 3. **Secondary Winding**
 4. **Insulation**
 5. **Cooling medium (oil or air)**
-

Function

- Core provides low reluctance path
 - Windings transfer electrical energy
-
-

3 EMF Equation of Single-Phase Transformer

Assumptions

- Sinusoidal flux
 - Negligible losses
 - Uniform flux distribution
-

Derivation

Let maximum flux = Φ_m

$$\Phi = \Phi_m \sin \omega t$$

Induced EMF:

$$e = N \frac{d\Phi}{dt}$$

$$e = N\omega\Phi_m \cos \omega t$$

Maximum EMF:

$$E_m = N\omega\Phi_m$$

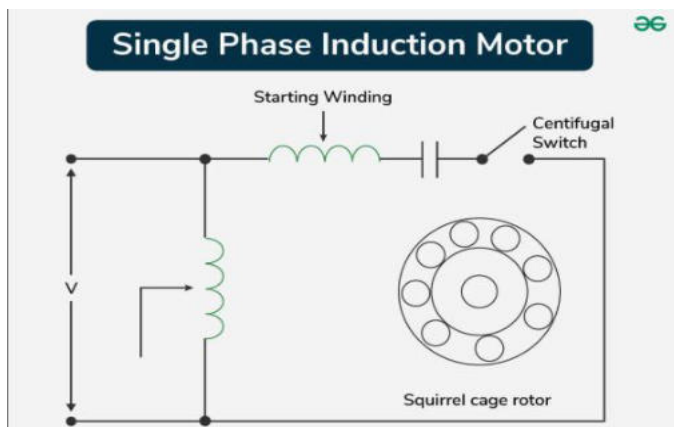
RMS EMF:

$$E = \frac{E_m}{\sqrt{2}} = 4.44fN\Phi_m$$

EMF Equations

$$E_1 = 4.44fN_1\Phi_m$$

$$E_2 = 4.44fN_2\Phi_m$$



Core Type vs Shell Type Transformers

Parameter	Core Type	Shell Type
Magnetic Circuit	Single	Double
Winding	Surrounds core	Surrounded by core

Parameter	Core Type	Shell Type
Leakage Flux	More	Less
Mechanical Strength	Less	More
Cooling	Better	Moderate

5 Auto Transformer

Definition

An auto transformer uses a **single winding** for both primary and secondary.

Advantages

1. Less copper required
 2. Smaller size
 3. Higher efficiency
 4. Better voltage regulation
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Disadvantages

1. No electrical isolation
 2. Higher fault current
 3. Not suitable for large voltage ratios
-

Applications

- Voltage regulators
 - Motor starters
 - Power transmission
-

6 3-Phase Transformer Connections

Common Connections

Connection	Features
Star–Star (Y–Y)	Neutral available
Delta–Delta (Δ – Δ)	Used for heavy loads
Delta–Star (Δ –Y)	Used in transmission
Star–Delta (Y– Δ)	Used in distribution

7 Ideal vs Practical Transformer

Parameter	Ideal	Practical
Losses	Zero	Copper & core losses
Efficiency	100%	<100%
Leakage Flux	Nil	Present
Regulation	Zero	Non-zero

Electrical Circuit vs Magnetic Circuit (Short)

Electrical Circuit	Magnetic Circuit
Current flows (I)	Flux flows (Φ)
Driven by EMF (V)	Driven by MMF (AT)
Opposed by Resistance (R)	Opposed by Reluctance ($\mathfrak{R}$)
Obeys Ohm's law	Obeys Hopkinson's law
Power loss as heat	Core losses occur

MODULE 4: FUNDAMENTALS OF ELECTRICAL MACHINES

4.1 THEORY QUESTIONS (IN DETAIL, EXAM-READY)

1 SINGLE-PHASE INDUCTION MOTOR

(a) Working Principle

Principle

A single-phase induction motor works on the principle of **electromagnetic induction**.

When a single-phase AC supply is given to the stator winding:

- An **alternating magnetic field** is produced
 - This induces EMF in the rotor conductors
 - Rotor current flows
 - Interaction of stator flux and rotor current produces torque
-

? Why Single-Phase Induction Motor is NOT Self-Starting

- Single-phase supply produces **pulsating magnetic field**, not rotating
- This field can be resolved into **two equal rotating fields** in opposite directions
- At standstill, torques produced by both fields are equal and opposite
- Net torque = **zero**

✓ Hence, motor **does not start on its own**

(b) Types of Single-Phase Induction Motors

1. Split Phase Induction Motor

Construction

- Main winding (low resistance, high inductance)
- Auxiliary winding (high resistance, low inductance)
- Centrifugal switch

Working

- Phase difference between currents in main and auxiliary winding
- Produces starting torque
- Auxiliary winding disconnected after starting

2. Capacitor Start Induction Motor

Construction

- Capacitor connected in series with auxiliary winding

Working

- Capacitor creates large phase difference ($\sim 90^\circ$)
- High starting torque obtained
- Capacitor disconnected after starting

3. Capacitor Start–Capacitor Run Motor

- One capacitor for starting
- One capacitor permanently connected
- Smooth operation and high efficiency

(c) Merits, Demerits & Applications

Merits

1. Simple construction
2. Low cost
3. Easy maintenance

Demerits

1. Low starting torque
 2. Low efficiency
 3. Not suitable for heavy loads
-

Applications

- Fans
 - Washing machines
 - Refrigerators
 - Pumps
-
-

THREE-PHASE INDUCTION MOTOR

(a) Generation of Rotating Magnetic Field (RMF)

Explanation

When a three-phase supply is applied to stator windings:

- Three currents displaced by 120° flow
 - Each current produces a magnetic flux
 - Resultant flux rotates at synchronous speed
-

Synchronous Speed

$$N_s = \frac{120f}{P}$$

Where

f = supply frequency

P = number of poles

Importance of RMF

- RMF cuts rotor conductors
 - Induces current in rotor
 - Produces torque
-

(b) Comparison: Squirrel Cage vs Slip Ring IM

Parameter	Squirrel Cage	Slip Ring
Rotor	Aluminium bars	Wound rotor
Starting torque	Low	High
Starting current	High	Low
Cost	Low	High
Maintenance	Very low	High
Application	Fans, pumps	Cranes, lifts

3 DC MACHINES

(a) Construction & Parts of DC Machine

Main Parts

1. **Yoke** – Outer frame, mechanical support
 2. **Pole core & pole shoe** – Spread flux
 3. **Field winding** – Produces magnetic field
 4. **Armature core** – Laminated, reduces losses
 5. **Armature winding** – Carries current
 6. **Commutator** – Converts AC to DC
 7. **Brushes** – Collect current
-

(b) Classification of DC Motors

Based on Excitation

1. Separately excited motor
 2. Shunt motor
 3. Series motor
 4. Compound motor
 - Cumulative
 - Differential
-

(c) Significance of Back EMF

Definition

Back EMF is the voltage induced in armature conductors due to rotation in magnetic field.

$$E_b = \frac{P\Phi ZN}{60A}$$

Importance

1. Limits armature current
 2. Protects motor from damage
 3. Controls speed automatically
 4. Indicates load condition
-

(d) Brushless DC Motor (BLDC)

Construction

- Permanent magnet rotor
- Stator with 3-phase windings
- Electronic controller (no brushes)

Working

- DC supply converted into AC by controller
- Sequential energizing of stator windings
- Rotor follows rotating magnetic field

Advantages

- High efficiency
- No sparking
- Low maintenance

Applications

- Electric vehicles
 - Drones
 - Fans
-
-

4 SYNCHRONOUS GENERATOR (ALTERNATOR)

Construction

Main Parts

1. **Rotor**
 - Salient pole
 - Cylindrical rotor
 2. **Stator** – 3-phase armature winding
 3. **Excitation system**
-

Working Principle

- DC excitation produces magnetic field on rotor
 - Rotor rotated by prime mover
 - Magnetic field cuts stator conductors
 - EMF induced based on Faraday's law
-

EMF Equation

$$E = 4.44f\Phi T$$

Applications

- Power stations
 - Diesel generators
 - Wind power plants
-

MODULE 5: ELECTRICAL WIRING, SAFETY & PROTECTION

5.1 THEORY QUESTIONS (IN DETAIL, EXAM-READY)

1 PROTECTIVE DEVICES

(a) Miniature Circuit Breaker (MCB)

Definition

MCB is an **automatic protective device** that protects electrical circuits from **overload and short circuit.**

Construction

- Bimetallic strip (thermal element)
 - Solenoid (magnetic element)
 - Contacts
 - Arc chute
 - Trip mechanism
-

Working

Overload Condition

- Excess current heats bimetal strip
- Strip bends and trips contacts
- Circuit opens

Short Circuit Condition

- Heavy current energizes solenoid
 - Instantaneous tripping
 - Circuit disconnected immediately
-

Advantages

- Fast operation
 - Resettable
 - More reliable than fuse
-

(b) Earth Leakage Circuit Breaker (ELCB)

Definition

ELCB disconnects the supply when **leakage current flows to earth**, protecting users from electric shock.

Working Principle

- Continuously monitors current difference between phase and neutral

- If leakage exceeds safe limit (≈ 30 mA)
- Circuit trips immediately

Application

- Homes
- Hospitals
- Wet areas

(c) Moulded Case Circuit Breaker (MCCB)

Definition

MCCB is a heavy-duty circuit breaker used for **high current and industrial applications**.

Features

- Adjustable trip settings
- Handles large current ratings
- High breaking capacity

Applications

- Factories
- Commercial buildings
- Motor protection

Difference Between Fuse and MCB

Parameter	Fuse	MCB
Operation	Melts wire	Trips mechanically
Reusability	Not reusable	Reusable
Speed	Slower	Faster

Parameter	Fuse	MCB
Cost	Low	Higher
Reliability	Less	More

(d) Switch Fuse Unit (SFU)

Definition

SFU is a **combined unit** consisting of a switch and fuse used for isolation and protection.

Function

- Manual switching
 - Protection against overload & short circuit
-
-

Application

- Distribution boards
 - Industrial panels
-
-

2 EARTHING

(a) Necessity of Earthing

Why Earthing is Required

1. Protects humans from electric shock
 2. Prevents damage to equipment
 3. Provides low resistance path for fault current
 4. Stabilizes system voltage
 5. Reduces fire hazards
-

(b) Methods of Earthing

1. Plate Earthing
2. Pipe Earthing

(c) Plate Earthing

Construction

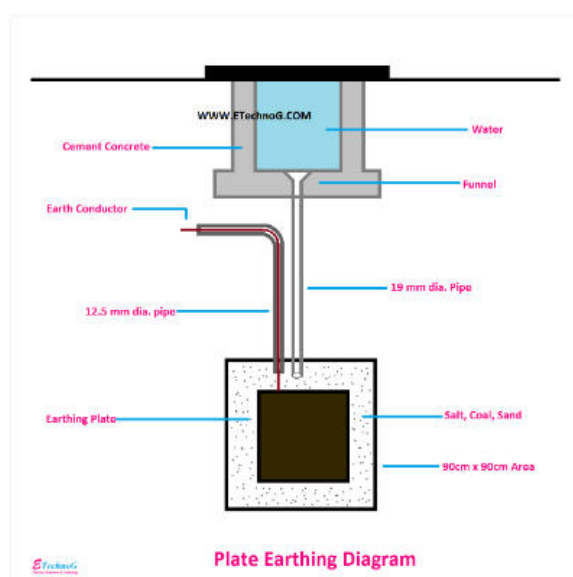
- Copper or GI plate buried vertically
- Plate size: 600 × 600 mm
- Surrounded by charcoal and salt
- Connected to equipment using earthing strip

Working

- Moist soil reduces resistance
- Fault current flows safely to earth

Applications

- Residential buildings
- Small installations



(d) Pipe Earthing

Construction

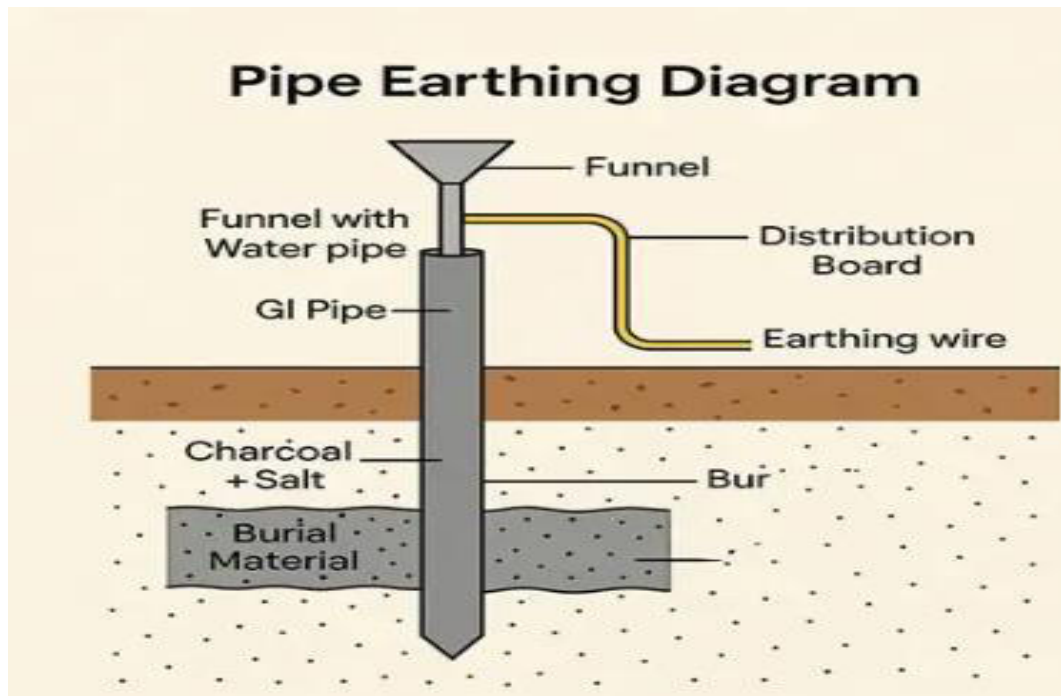
- GI pipe buried vertically
 - Length: 2.5–3 meters
 - Charcoal and salt packed around pipe
-

Advantages

- Low earth resistance
 - Economical
 - Most commonly used
-

Applications

- Industrial installations
- Power systems



3 SAFETY

(a) Safety Precautions Against Electrical Shock

1. Proper earthing
 2. Use of MCB and ELCB
 3. Avoid wet hands while operating switches
 4. Insulated tools
 5. Periodic maintenance
 6. Proper insulation of wires
-

(b) Safety Devices for Home Appliances

1. Fuse
2. MCB

3. ELCB / RCCB
 4. Earthing wire
 5. Insulated plugs and sockets
 6. Voltage stabilizer
-

4 WIRING & CABLES

Structure of Underground Cable

Main Layers

1. **Conductor**
 - Copper or aluminium
 - Carries current
2. **Insulation**
 - PVC / XLPE
 - Prevents leakage
3. **Metallic Sheath**
 - Lead or aluminium
 - Moisture protection
4. **Bedding**
 - Jute or PVC
 - Mechanical protection
5. **Armouring**
 - Steel wire/tape
 - Protects from damage
6. **Sheath Outer**
 - PVC
 - Environmental protection

Advantages of Underground Cables

- Safe
 - Less maintenance
 - No visual pollution
-

MODULE 5: ELECTRICAL WIRING, SAFETY & PROTECTION

(Remaining Theory Questions – In Detail)

1 Difference between Wire and Cable (S-23)

Parameter	Wire	Cable
Definition	Single conductor	Two or more conductors
Insulation	May or may not be insulated	Always insulated
Protection	Less mechanical protection	More protection (sheath, armour)
Flexibility	Less	More
Application	Internal wiring	Power transmission & distribution

2 Batteries

(a) Types of Batteries

1. Primary Batteries

- Cannot be recharged
- Chemical reaction irreversible
- Example: Dry cell

2. Secondary Batteries

- Rechargeable
- Chemical reaction reversible

- Example: Lead-acid battery
-

(b) Characteristics of a Battery

1. **Nominal Voltage** – Rated output voltage
 2. **Capacity (Ah)** – Amount of charge supplied
 3. **Efficiency** – Ratio of output to input energy
 4. **Internal Resistance** – Opposition within battery
 5. **Life** – Number of charge–discharge cycles
-

(c) Construction of Lead Acid Battery

Main Parts

1. **Container**
 - Acid-resistant plastic
 2. **Positive Plate**
 - Lead dioxide (PbO_2)
 3. **Negative Plate**
 - Spongy lead (Pb)
 4. **Electrolyte**
 - Dilute sulphuric acid (H_2SO_4)
 5. **Separator**
 - Insulating material between plates
 6. **Vent Plug**
 - Releases gases
-

Working (Brief)

- During discharge → chemical energy → electrical energy
 - During charging → electrical energy → chemical energy
-

Applications

- Automobiles
 - UPS
 - Power backup systems
-

3 Illumination

(a) Types of Lamps

1. Incandescent Lamp

- Filament heated to glow
- Low efficiency

2. Fluorescent Lamp

- Gas discharge type
- Higher efficiency

3. LED Lamp

- Semiconductor based
 - Very high efficiency
 - Long life
-

(b) Lumen Requirement

Lumen

Unit of **luminous flux**.

Importance

- Determines brightness of area

- Depends on:
 - Area size
 - Type of work
 - Height of mounting
-

Typical Lumen Levels

- Study room: High lumen
 - Corridor: Low lumen
 - Workshop: Very high lumen
-

4 Power Factor Improvement

(a) Need for Power Factor Improvement

1. Reduces power loss
 2. Improves system efficiency
 3. Reduces electricity bill
 4. Increases load capacity
 5. Improves voltage regulation
-

(b) Methods of Power Factor Improvement

1. Static Capacitors

- Connected across load
 - Supply leading reactive power
-

2. Synchronous Condenser

- Over-excited synchronous motor
 - Adjustable power factor
-

3. Phase Advancers

- Used in induction motors
 - Supplies magnetizing current
-

5 Energy Meter

Working Principle of Energy Meter

Type

Single-phase **Induction Type Energy Meter**

Principle

Works on the principle of **electromagnetic induction**.

Main Parts

1. Current coil
 2. Pressure coil
 3. Aluminium disc
 4. Permanent magnet
 5. Registering mechanism
-
-

Working

- Current coil produces flux proportional to load current
- Pressure coil produces flux proportional to voltage
- Interaction of fluxes produces torque
- Aluminium disc rotates
- Speed proportional to power
- Number of revolutions proportional to energy consumed

$$\text{Energy} = \text{Power} \times \text{Time}$$

Advantages

- Simple construction
- Reliable
- Accurate

• Comparison between MCB and ELCB

Basis	MCB (Miniature Circuit Breaker)	ELCB (Earth Leakage Circuit Breaker)
Full form	Miniature Circuit Breaker	Earth Leakage Circuit Breaker
Purpose	Protects circuit from overload and short circuit	Protects against earth leakage current
Type of protection	Overcurrent protection	Shock and fire protection
Working principle	Thermal (bimetal strip) and magnetic mechanism	Detects leakage current flowing to earth